

STA138 Final Project

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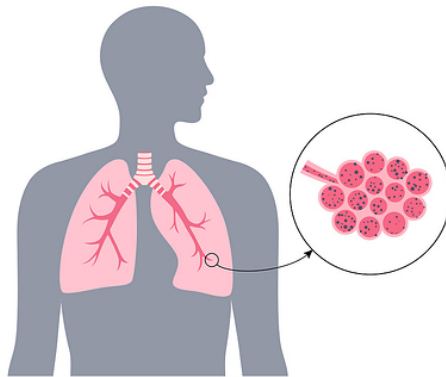
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I. Introduction

Byssinosis is an occupational respiratory disease caused by inhaling cotton or dust in inadequately ventilated working places, and it can worsen over time with repeated exposure. Byssinosis is frequently seen in textile workers in the yarn and fabric manufacture business. Byssinosis has some symptoms such as chest tightness, coughing, and wheezing, particularly at the beginning of the workweek. Byssinosis often occurs in the periods of the Industrial Revolution. Between 1996 to 2005, North Carolina accounted for about 37% of all byssinosis-related deaths in the United States ([WIKIPEDIA](#)).

In 1973, a comprehensive study was conducted involving 5,419 workers from a large cotton textile company in North Carolina. The aim was to determine the extent to which factors including sex, race, length of employment, smoking habit and dustiness of the workplace account for byssinosis. This project was desired to discover the association between byssinosis and these variables by applying exploratory data analysis and appropriate statistical modeling. We performed the forward stepwise regression to select the model that explains the data well and we used the AIC value as criteria to compare between models. A particular focus will be determining whether workplace dustiness significantly contributes to the likelihood of developing byssinosis, which we used the Likelihood Ratio Test to test the null hypothesis with significance level 0.05.

II. Exploratory Data Analysis (EDA)

The dataset contains observations from 5419 workers that includes 5 categorical predictors such as the type of work place, years of employment, smoking status, sex, and race. The response variable is binary, where it indicates the number of workers who suffer from byssinosis or not. There are 72 possible combinations from these variables. From the summary of the data, 31 workers out of 5419

workers (0.57%) suffer from byssinosis, which is a relatively low rate of byssinosis diagnosis.

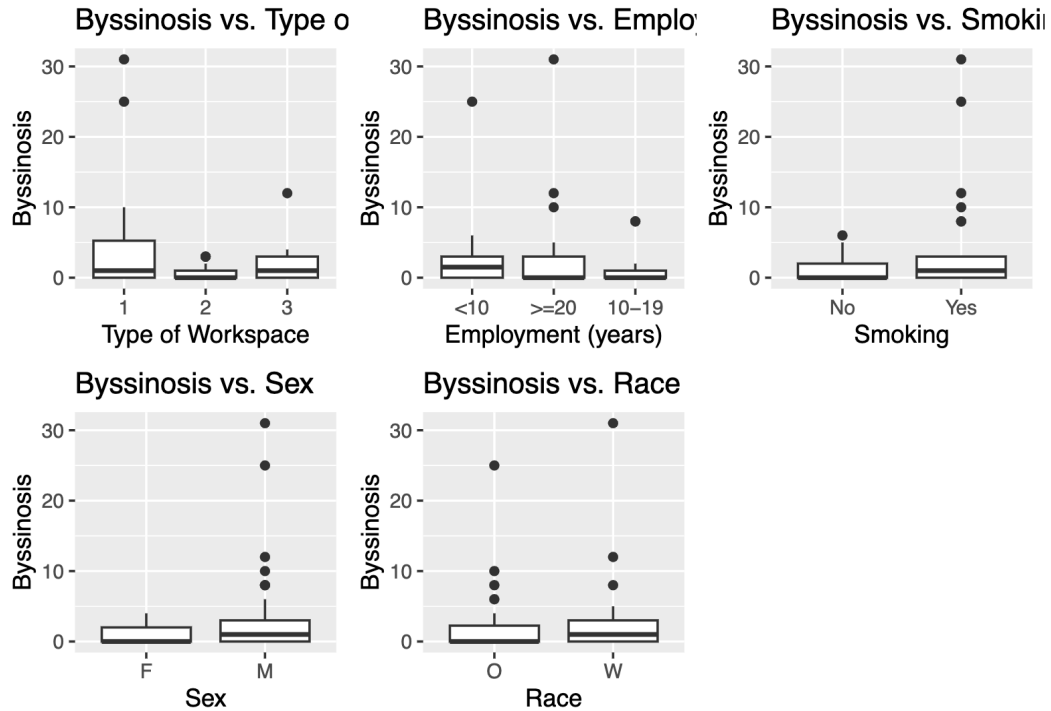


Figure 1 - Boxplots of Byssinosis vs. Other variables

From the boxplots, we can see that workers in the most dusty workspace (Type 1) have a significantly higher byssinosis level than the other two workspace. When comparing between employment periods, the longer employment (> 20 years) seems to have higher byssinosis levels. There is a higher level of having byssinosis for smokers when comparing smoking status. For different genders, there is a slight trend towards male having more cases of byssinosis. There is not much difference in byssinosis levels across different races. This might suggest that race and sex might not be the significant factors that influence the chance of having byssinosis. Additionally, we converted the workspace variable into a factor variable to ensure a more accurate result in the logistic regression model.

III. Model Selection and Inference

Since we have to use our data for both model fitting and inference, we would have to split our data into 90% training data and 10% testing data to prevent overfitting that might lead to inaccurate results.

1. Inference - Whether workspace dustiness is a risk factor?

The risk factor means that there is some association of log odds of byssinosis and the workspace variable. Since we want to test whether workspace is a significant factor in predicting the chance of byssinosis, we would use the Likelihood Ratio Test instead of the Wald test to test for the significance of the parameter because it has 3 distinct levels and the Wald test might give us contradicting results among these 3 levels.

First, we used forward stepwise regression to identify the best model based on AIC using the training data, and we searched only within models that include the workspace variable. We obtained a null model that includes workspace, smoking, employment, and the interaction term between workspace and smoking. Then, we set the alternative model that omits the workspace variable, which left us with smoking and employment variables. Using the likelihood ratio test, we computed the test statistics using the formula $-2(L_1 - L_0)$, where L_1 and L_0 are the log likelihood of the null and alternative models respectively. Then, the p-value calculated under the null chi-squared distribution with degrees of freedom of 1 is 0.04764, which is less than the significance level 0.05. Since the p-value is less than the significance level, we reject the null hypothesis and conclude that there is significant evidence that workspace is a significant predictor (association) for byssinosis.

2. Model Fitting

To select the model that best predicts the chance of having byssinosis, we performed the forward stepwise regression on the training data beginning with the null model with no predictors, and increasing the complexity of our model by adding in predictors and interaction terms. Then, we selected the model with the lowest AIC or BIC value, which means that the model explains the data well without overfitting. The AIC selected model is as follows:

```
##
## Call:  glm(formula = cbind(Byssinosis, Non.Byssinosis) ~ Workspace +
##      Employment + Smoking + Workspace:Smoking, family = binomial,
##      data = byssinosisTrain)
##
## Coefficients:
##      (Intercept)          Workspace2          Workspace3
##          -2.7320           -1.5761           -2.2994
##      Employment>=20      Employment10-19      SmokingYes
##          0.6948           0.5292           0.9633
## Workspace2:SmokingYes  Workspace3:SmokingYes
##          -1.3815           -0.5423
##
## Degrees of Freedom: 58 Total (i.e. Null);  51 Residual
## Null Deviance:      282.8
## Residual Deviance: 35.34    AIC: 146.7
```

The AIC selected model includes the workspace, employment, smoking variables, and the interaction term between workspace and smoking.

The following is for the BIC selected model:

```
##
## Call:  glm(formula = cbind(Byssinosis, Non.Byssinosis) ~ Workspace +
##      Smoking + Employment, family = binomial, data = byssinosisTrain)
##
## Coefficients:
##      (Intercept)          Workspace2          Workspace3      SmokingYes
##          -2.4417           -2.5176           -2.7003           0.6163
##      Employment>=20      Employment10-19
##          0.6399           0.5252
##
## Degrees of Freedom: 58 Total (i.e. Null);  53 Residual
## Null Deviance:      282.8
## Residual Deviance: 40.34    AIC: 147.7
```

The BIC selected model includes the workspace, employment, smoking variables, but without the interaction term. The BIC selected model tends to choose the simpler model because it gives more penalty on the sample size and the number of parameters. From the AIC selected model, the AIC value is 146.7. In contrast, the BIC value is 147.7 from the BIC selected model. Therefore, we will choose the AIC selected model with the workspace, employment, smoking variables, and the interaction term between workspace and smoking. The following output includes the coefficients and their significance for the AIC selected model:

```
##
## Call:
## glm(formula = cbind(Byssinosis, Non.Byssinosis) ~ Workspace +
##      Employment + Smoking + Workspace:Smoking, family = binomial,
##      data = byssinosisTrain)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -2.7320     0.2728  -10.014 < 2e-16 ***
## Workspace2        -1.5761     0.4870   -3.236 0.001212 **
## Workspace3        -2.2994     0.3777   -6.087 1.15e-09 ***
## Employment>=20     0.6948     0.1948    3.566 0.000362 ***
## Employment10-19    0.5292     0.2567    2.061 0.039265 *
## SmokingYes         0.9633     0.2865    3.363 0.000772 ***
## Workspace2:SmokingYes -1.3815     0.6366   -2.170 0.029989 *
## Workspace3:SmokingYes -0.5423     0.4403   -1.232 0.218080
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 282.767  on 58  degrees of freedom
## Residual deviance:  35.336  on 51  degrees of freedom
## AIC: 146.66
##
## Number of Fisher Scoring iterations: 5
```

IV. Model Interpretation and Model Diagnostics

1. Model Interpretation

Using the coefficients found in our analysis above, the formula for the final model is as follows:

$$\log\left(\frac{\pi}{1-\pi}\right) = -2.7320 - 1.5761x_{\text{workspace,lessDusty}} - 2.2994x_{\text{workspace,leastDusty}} + 0.6948x_{\text{employment,20}} + 0.5292x_{\text{employment,10-19}} + 0.9633x_{\text{smoking,yes}} - 1.3815x_{\text{workspace,lessDusty:smoking,yes}}$$

We can interpret the coefficients of the model in terms of their effect on the odds of the status of byssinosis by using the coefficient values above. Values for employment for years greater than or equal to 10, as well as positive smoking status increase the chance of having byssinosis; concurrently, values for the type of workspace (both less and least dusty environments), as well an interaction effect between workspace (less dusty) and smoking status (yes smoking) result in a decrease for the chance of having byssinosis, all else equal. The interaction term for least dusty workspace and positive smoking status was excluded as it was not statistically significant ($p = .218$). Excluding this variable ensures model simplicity and interpretability.

2. Model Diagnostics

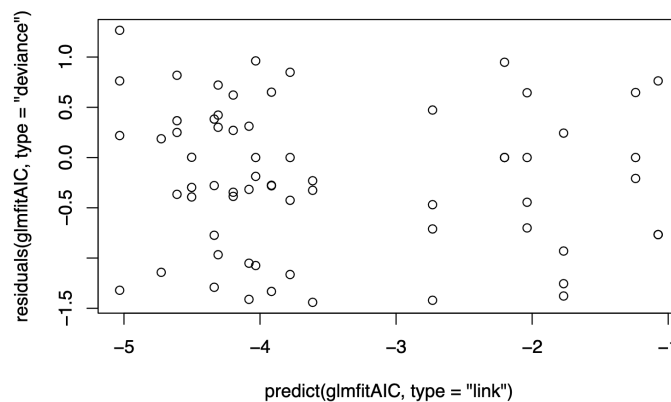


Figure 2 - Residual Plot of AIC Selected Model

This plot is the residuals vs. predicted values plot, displaying deviance residuals on the y-axis against predicted values (log-odds) on the x-axis for our model. The

residuals are randomly distributed around zero, indicating that the model does not overestimate or underestimate the outcomes. The spread of the residuals appears consistent, suggesting it is a good overall fit without evident heteroscedasticity.

V. Conclusion

In conclusion, we were able to speculate that the workspace dustiness was potentially a significant variable from looking at the exploratory data analysis box plots. We proceeded with the model selection process using the forward stepwise regression based on the split 90% training data and searched only within models that included the workspace variable. From selecting a model that gave us the lowest AIC, we obtained the null model that had workspace, smoking, employment, and the interaction term between workspace and smoking variables. After this we obtained an alternative model that omits the workspace variable, leaving the model with the smoking and employment variables. We performed the likelihood ratio test and computed the test statistic. The p-value is calculated under the null chi-squared distribution with degrees of freedom of 1 and is 0.04764. Our alpha value is 0.05, meaning that our p-value is smaller than our significance level. Therefore, we reject the null hypothesis and conclude that we have sufficient evidence that workspace and byssinosis are associated. After this, we do model fitting to obtain the best model that predicts the chance of byssinosis best. We obtain the model with the lowest AIC value which is the model that includes workspace, employment, smoking variables, and the interaction term between workspace and smoking. This model reveals that the longer years of employment (10 or more) and smoking are associated with an increased chance of developing byssinosis. On the other hand, working spaces that are less dusty and the interaction between the less dusty working spaces and smoking reduce the likelihood of byssinosis when we assume that the other factors remain constant.